YouTube Video View Prediction and Feature Importance Analysis

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Overview

- Developed a Random Forest Regression model for predicting YouTube music-related video view, achieving a test R^2 score of approximately 0.71 and a test score of 0.61.
- Analyzed feature importance and visualized key factors influencing video views to understand the direction of correlation.
- Processed and cleaned the collected data, applied proper transformation and encodings to improve model performance and interpretability.

1. Introduction

Inspired by the project idea of analyzing what makes a Reddit submission good or bad, this study will be analyzing YouTube music videos. Accurately predicting video performance allows content creators to optimize their publishing strategies and improve audience engagement. It also helps reduce situations where good content fails to gain more views due to unfavorable timing or lack of alignment with the YouTube algorithms. By understanding which factors impact view counts, it allows creators to maximize the chance of gaining exposure by making wise decisions on how to release their videos.

1.1. Problem Description

As the internet continues to evolve, Youtube has become one of the most influential platforms for sharing content to worldwide audiences. Currently, there are billions of videos available on Youtube, and music-related videos often occur a significant portion of the trending list. However, predicting the number of views a new video will receive or identifying the factors that influence view counts is challenging. One major question is whether there are factors beyond simply how “good” a song is, that can affect the view counts. Therefore, the objective of this project is to identify the key factors influencing the popularity of music video and predict the number of views a YouTube video will receive within a specified week after publishing.

1.2. Report Structure

The report is divided as follows: Section 2 - Data Description, describes all input elements used for this analyzation, including the dataset information and the data schema. Section 3 - Method, covers the approach used for this project, including the rationale on data cleaning and analytical techniques. Section 4 - Results, discuss the result from analysis. Section 5 - Future Work, provides some suggestions for future improvements. Lastly, section 6 - Conclusion, summarized the work done and addressed the answer to the project question.

2. Data Description

The dataset contains publicly available statistics for YouTube videos, including each video’s channel subscriber count, the duration, publication date, and engagement measures such as the number of views, likes, comments for each music-related video. These statistical data were collected using custom Python scripts named in the format *get*.py* (e.g. *getEN.py*). In addition to the numerical data, several categorical data such as the singer’s style (e.g. JPop), the type of music video (e.g. original release, song cover), the week of day, and the season of releasing date were added to the dataset for enabling more board analysis. Since the YouTube API does not directly provide such categorical metadata, these fields were derived from the statistical information. While the date-related attributes are straightforward to compute, determining the music type is more challenging and was derived from searching key words in the video title. Since we are only interested in music-related videos, the non-relative videos were skipped by filtering out unwanted types. Because different channels have different naming patterns, the data are collected individually for each channel before merging the datasets, ensuring consistency in classifying the type.

As view counts can change every minute, the data were collected within the same night to maintain consistency. The processed data was then exported as a CSV file, with a ready-to-run version that contains all chosen channel's historical video records up to August 9, 2025.

3. Method

3.1. Data Processing

Convert As row-level cleaning was performed during the data collection stage, the processing in the main script focuses on column-level adjustments. To prepare the data for the machine learning model, categorical values are converted into standardized rank values to ensure the model can process all features numerically. The ranks are determined by which category value has the highest average view count, ensuring that more influential category values receive higher ranks.

Drop To improve the data quality, unhelpful columns are dropped from the dataset to reduce noise. This includes the channelId because it has no predictive meaning besides identifying the channel, and users would not know their channel ID when using the prediction function. Similarly, likes, comments, and engagement are excluded since they would all be zero for a new published video during the predicting stage which are not useful to the model. Then the title is also dropped because it requires natural language processing for converting into useful numerical value.

Add To address the fact that older videos get more time to accumulate views than more recent videos, a new column is added to indicate the number of weeks since the video's release. This is to provide the model to incorporate weeks as a predictive factor, learning how view counts change over different time spans.

Transform Since the view is heavily right-skewed, it is log-transformed using the log1p function to improve interpretability and avoid issues when the view is zero.

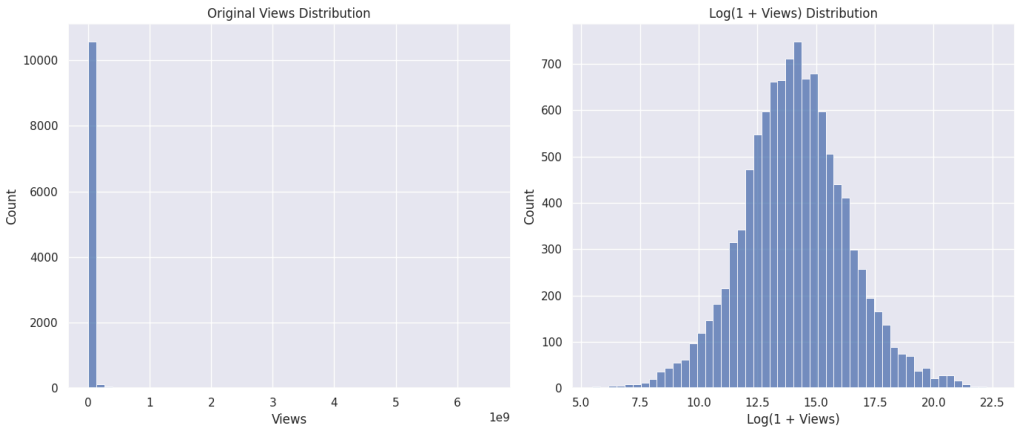

Fig.1: views distribution histogram

3.2. Analytical Techniques

Split Data Since the goal is to predict video views, the target variable is the views column, and so the remaining columns serve as features. Then, the dataset is split into two subsets, for training and testing the model.

Training A Random Forest Regressor was used in this analysis due to its strength in handling non-linear relationships, the ability to capture complex interactions between multiple features, and reduces the risk of over-fitting by averaging predictions across different decision trees. The model performance is evaluated using the coefficient of determination (R^2 score). In this project, the Random Forest is chosen with 12 trees, a maximum depth of 25, and a minimum of 10 samples per leaf.

Importance After training, the most influential factors are identified using the model's feature_importances_ attribute, providing an answer to the question of which features have the most impact on video views.

4. Results

The model achieved a training score of approximately 0.71 and a test score of 0.61, indicating a good fit to the training data and reasonable prediction to unseen data.

Feature importance analysis shows that channel subscriptions are the most significant factor, with an importance score of 0.55. Followed by the number of weeks since the video was published, with an importance score of 0.20.

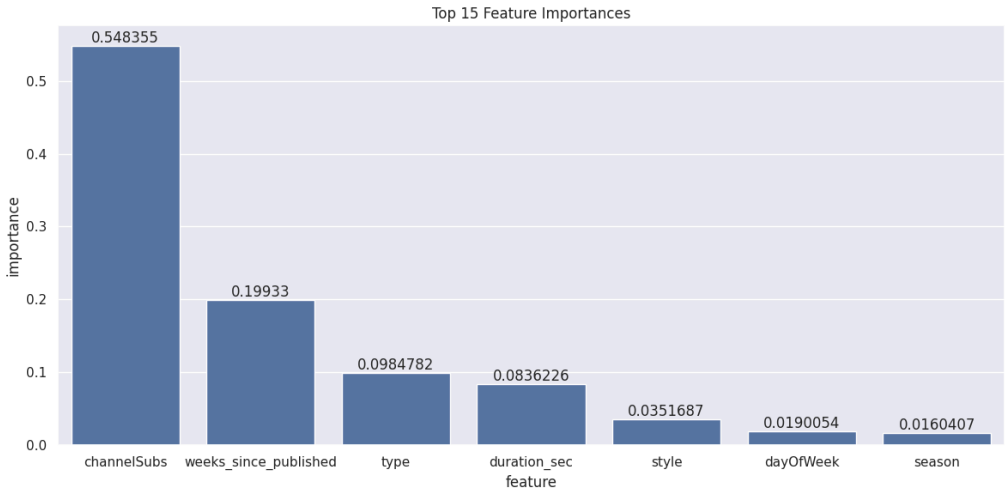

Fig. 2 Importance bar plot

Since the importance only shows the magnitude of impact but not the direction., scatter plots were used for understanding the direction of how the top three important features each influence views.

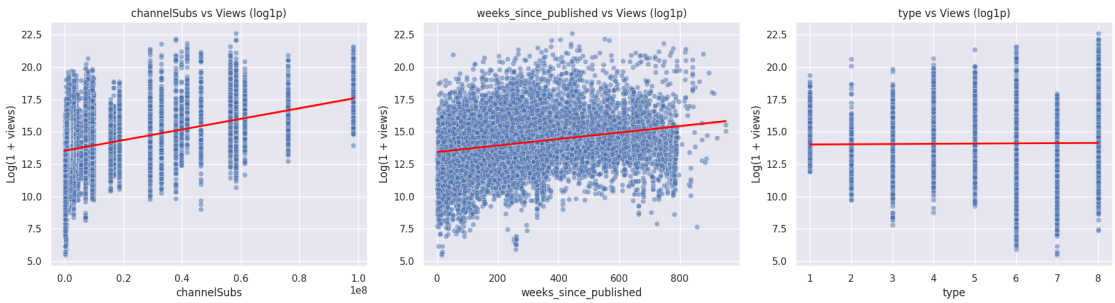

Fig. 3 top 3 important feature correlation scatter plots

The scatter plots suggest that only channel subscriptions have a clear upward trend with views. The relationship with weeks is less obvious and can be explained by the fact that older videos get more opportunities to accumulate views, so it is not considered. The video type seems to have no influence on view as the best-fit line has a nearly flat slope.

To further validate the effect of subscriptions, instead of plotting the correlation on all views, a plot of subscriber count versus average views per channel was created. With the scatter plot also showing an obvious upward trend, indicating that a larger subscriber base strongly correlates with higher video views.

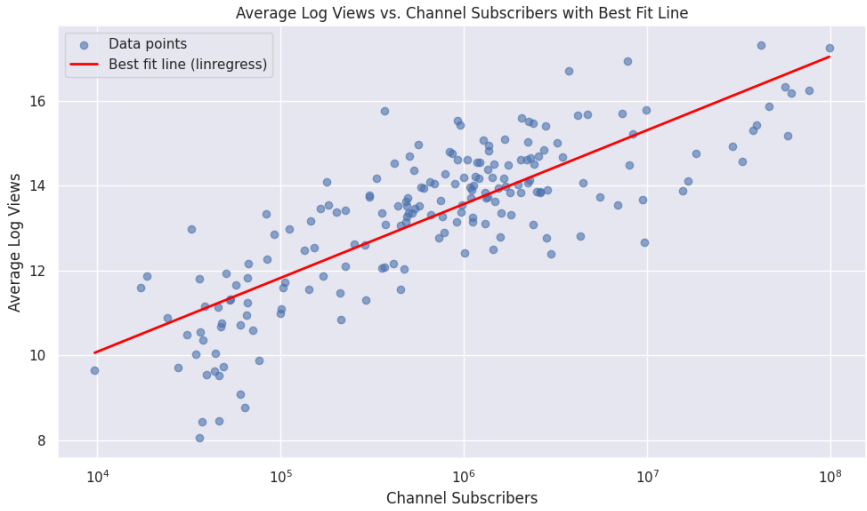

Fig 4. subscriber scatter plot

Therefore, the analysis suggests that the singer or the channel's popularity is the primary factor influencing the video popularity. Other features do not appear to have a significant impact on the view count.

5. Future Work

While the project only suggests an obvious finding that higher subscription leads to an overall higher view in music-related videos, there are likely other influential factors not captured in this study. Since the dataset is limited to metadata features, content-based features such as video title, thumbnail image or having more detailed content categories might enable the model to have better training and testing score, and generate more meaningful insights.

Additionally, the usability of the tool could be improved by developing an user interface for allowing users to input video details more efficiently without needing to modify variables in the program file.

6. Conclusion

The project analyzed music-related YouTube videos and used a Random Forest Regressor for forecasting view counts. The feature importance analysis shows that the channel subscription amount is the main factor that impacts the view count showing a strong positive correlation.